Week 1 Assignment: Data Exploration

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Course: ANA 505 Foundations of Data Mining

1. Dataset Preparation

The dataset `customer_data_week1_updated.csv` containing 100 customer records and 8 variables was successfully loaded and structured.

Data Structure (Variables & Types):

Variable	Data Type	Classification
CustomerID	int64 (Numeric)	Identifier
Age	int64 (Numeric)	Numeric
Gender	character (String)	Categorical
AnnualIncome	int64 (Numeric)	Numeric
MembershipYears	int64 (Numeric)	Numeric
TotalPurchases	int64 (Numeric)	Numeric
LastPurchaseAmount	float64 (Numeric)	Numeric
Customer_Segment	character (String)	Categorical

2. Descriptive Summary

There are no missing values across the 100 records in the dataset.

Summary Statistics (Numerical):

Statistic	Age	Annual Income	Membership Years	Total Purchases	Last Purchase Amount
Min	18.0	$30,604	1.0	1.0	$26.32

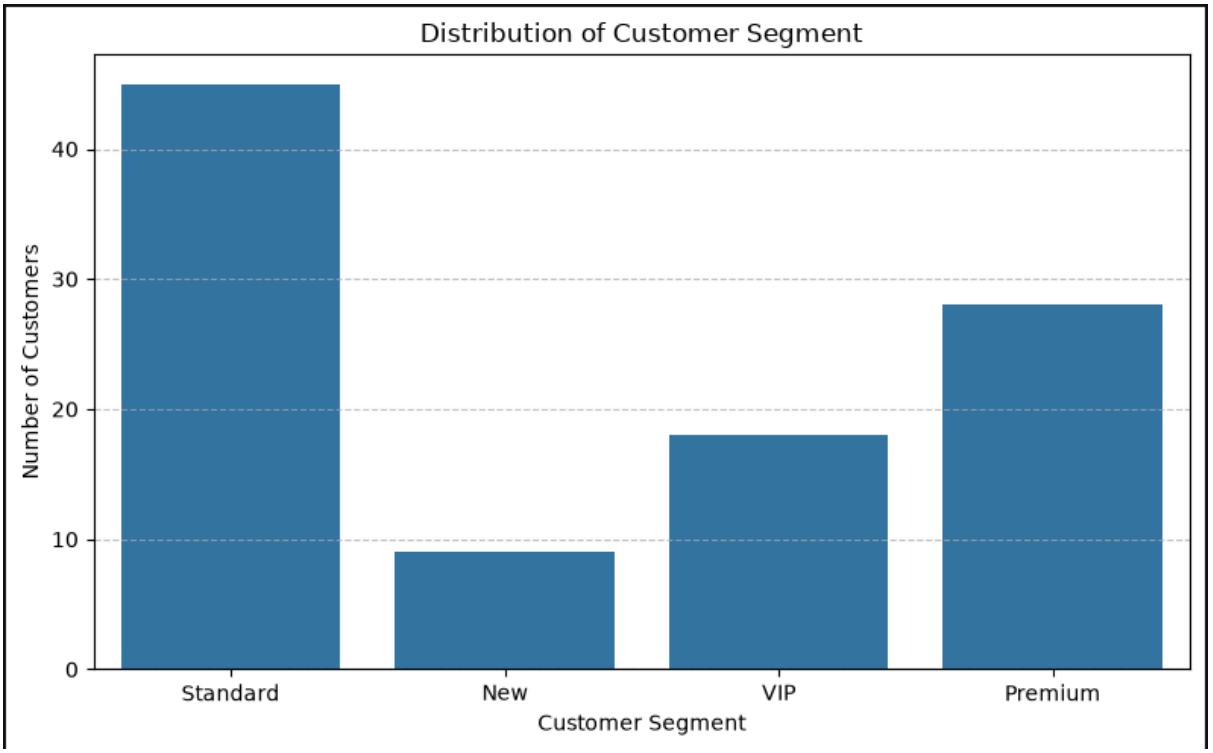
Statistic	Age	Annual Income	Membership Years	Total Purchases	Last Purchase Amount
Mean	44.38	$75,395.30	4.84	25.77	$245.02
Median	45.0	$73,340.00	4.0	24.5	$216.55
Max	69.0	$119,488	9.0	48.0	$499.48
Std Dev	15.40	$25,816.13	2.75	13.75	$143.17

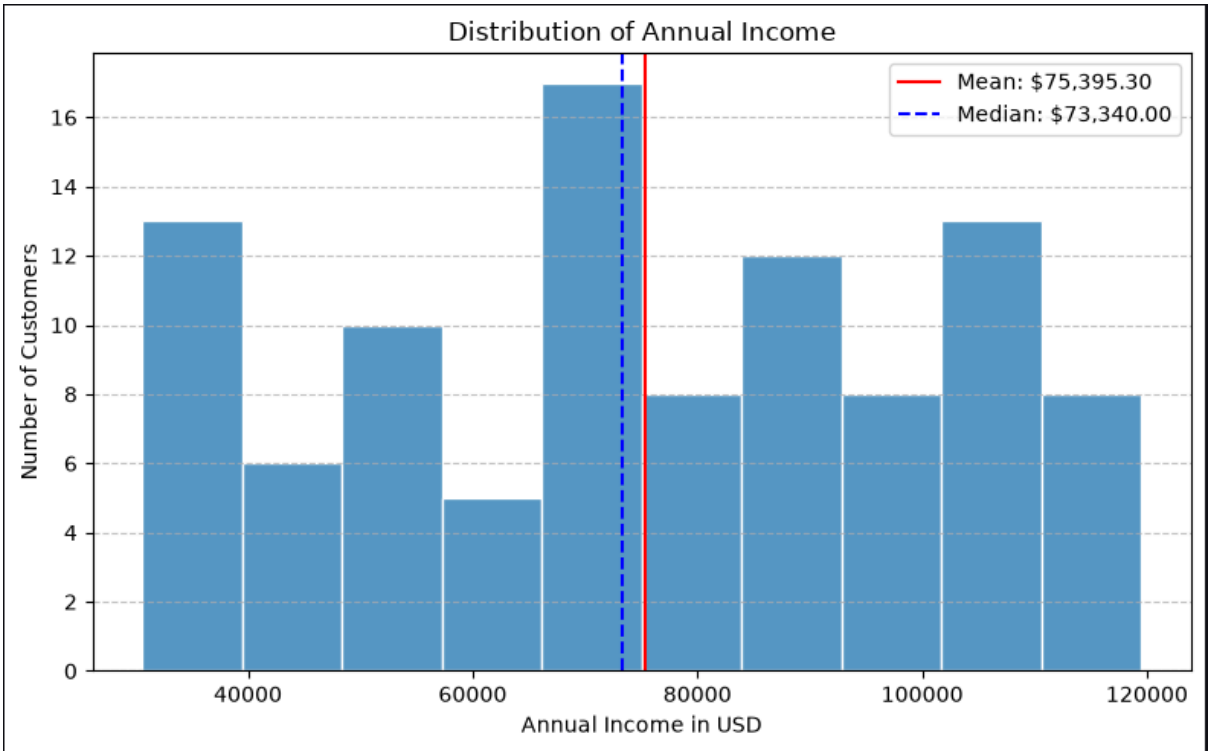
Customer Segment Distribution:

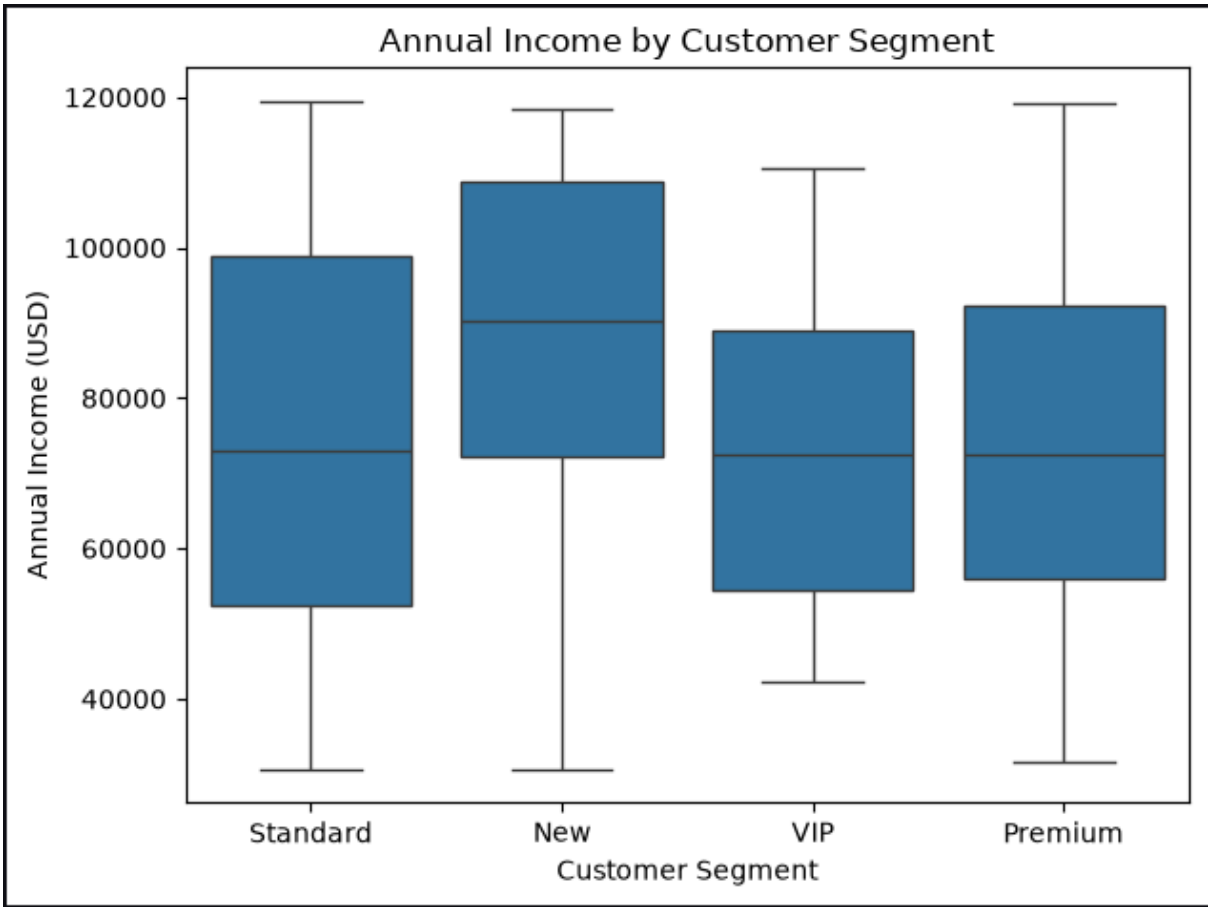
Segment	Count	Percentage
Standard	45	45.0%
Premium	28	28.0%
VIP	18	18.0%
New	9	9.0%

3. Basic Data Visualization

The visualizations are from the python script




Distribution of Annual Income
Mean: $75,395.30
Median: $73,340.00
Number of Customers
0
2
4
6
8
10
12
14
16
40000
60000
80000
100000
120000
Annual Income in USD


Annual Income by Customer Segment
Annual Income (USD)
120000
100000
80000
60000
40000
Standard
New
VIP
Premium
Customer Segment

4. Reflection

The dataset contains 100 customers and 8 variables describing their demographics, income, purchasing behaviour and customer segments. The variables include numerical measurements, namely, age, annual income, membership years and total purchases, as well as categorical variables, namely, gender, and customer segment. No missing values were identified.

The customer segment distribution shows the standard customer represents the largest group, accounting for 45% of the sample, followed by premium customers at 28%, VIP at 18% and new customers at 9%.

Annual income ranges approximately between $30.6k and $119.5k USD, with a mean income of $75.4k and median of $73.3k. An interesting observation is that the new customers have a higher average income of approximately $81.9k compared with other segments whose average incomes are relatively similar (averaging around $74.5k). However, the new segment contains only nine customers.

These variables could be useful for data mining since they provide information about customer characteristics and purchasing behaviour. Regression could be used to predict purchasing amounts based on age and income, while clustering could help identify hidden groups of customers with similar characteristics. Predictive classification could use variables like age, annual income, and total purchases to predict a customer's likelihood of upgrading from the standard to the VIP segment. These analyses could support customer profiling and more targeted, informed business decisions.

5. Appendix: R Script

Python

```
# Install required packages if they are not already installed
required_packages <- c("ggplot2", "dplyr")
missing_packages <- required_packages[
  !sapply(X = required_packages, FUN = requireNamespace, quietly = TRUE)
]
if (length(missing_packages) > 0) {
  install.packages(missing_packages, dependencies = TRUE)
}
library(ggplot2)
library(dplyr)
#============================================================
## 1. Read the dataset
#============================================================
fname <- "../datasets/customer_data_week1_updated.csv"
if (!file.exists(fname)) {
  stop(
```

```
    "Dataset not found. Update file_path so it points to
datasets/penguins_size.csv."
  )
}
customer_data <- read.csv(fname, stringsAsFactors = FALSE)
# Display the first few rows
head(customer_data)
#============================================================
## 2. Descriptive Summary
#============================================================
# Summary statistics for all variables
summary(customer_data)
# Summary statistics for numerical variables
numerical_data <- customer_data[
  c("Age", "AnnualIncome", "MembershipYears",
    "TotalPurchases", "LastPurchaseAmount")
]
summary(numerical_data)
# Standard deviations
sapply(numerical_data, sd, na.rm = TRUE)
# Display column names
names(customer_data)
# Display variable types
str(customer_data)
# Inspect unique customer segment values
unique(customer_data$Customer_Segment)
# Count customers in each segment
table(customer_data$Customer_Segment)
#============================================================
## 3. Basic Data visualization
#============================================================
# Missing values in each column
colSums(is.na(customer_data))
# Total number of missing values
sum(is.na(customer_data))
# Display rows containing at least one missing value
customer_data[
  !complete.cases(customer_data),
]
## 3.1 Bar chart: Customer segment distribution
# Count customers by segment
customer_segment_counts <- table(customer_data$Customer_Segment)
# Calculate percentages
customer_segment_percentages <- prop.table(
```

```r
  customer_segment_counts
) * 100
# Create summary table
segment_summary <- data.frame(
  Customer_Segment = names(customer_segment_counts),
  Count = as.integer(customer_segment_counts),
  Percentage = round(
    as.numeric(customer_segment_percentages),
    1
  )
)
segment_summary
# Bar chart
ggplot(customer_data, aes(x = Customer_Segment)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Distribution of Customer Segment",
    x = "Customer Segment",
    y = "Number of Customers"
  ) +
  theme_minimal()
## 3.2 Histogram: Annual income
# Calculate mean and median income
mean_income <- mean(
  customer_data$AnnualIncome,
  na.rm = TRUE
)
median_income <- median(
  customer_data$AnnualIncome,
  na.rm = TRUE
)
# Histogram
ggplot(customer_data, aes(x = AnnualIncome)) +
  geom_histogram(
    bins = 10,
    color = "white",
    fill = "steelblue"
  ) +
  geom_vline(
    xintercept = mean_income,
    color = "red",
    linetype = "solid"
  ) +
  geom_vline(
```

```r
    xintercept = median_income,
    color = "blue",
    linetype = "dashed"
  ) +
  labs(
    title = "Distribution of Annual Income",
    x = "Annual Income in USD",
    y = "Number of Customers"
  ) +
  theme_minimal()
#============================================================
## more analysis for reflection
#============================================================
# Annual income summary by segment
seg_income <- customer_data %>%
  group_by(Customer_Segment) %>%
  summarise(
    Count = n(),
    Mean = round(mean(AnnualIncome, na.rm = TRUE), 0),
    Median = round(median(AnnualIncome, na.rm = TRUE), 0)
  )
seg_income
# Age summary by segment
seg_age <- customer_data %>%
  group_by(Customer_Segment) %>%
  summarise(
    Count = n(),
    Mean = round(mean(Age, na.rm = TRUE), 0),
    Median = round(median(Age, na.rm = TRUE), 0)
  )
seg_age
ggplot(
  customer_data,
  aes(
    x = Customer_Segment,
    y = AnnualIncome
  )
) +
  geom_boxplot(fill = "steelblue") +
  labs(
    title = "Annual Income by Customer Segment",
    x = "Customer Segment",
    y = "Annual Income (USD)"
  ) +
```

```
theme_minimal()
```

6. Appendix: GitHub Repository

Python Script with Visualization:
https://github.com/m-abdelkawy/data-analysis-msc/blob/main/courses/ana505_data_mining/module1_intro/week1_intro/python/week1_activity.ipynb
R Script:
https://github.com/m-abdelkawy/data-analysis-msc/blob/main/courses/ana505_data_mining/module1_intro/week1_intro/r/week1_activity.R